Structure:

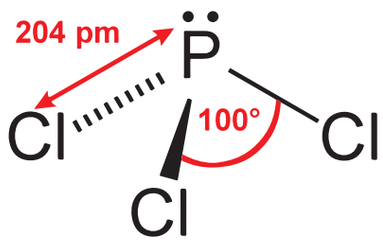


Figure 3.5 Structure of phosphorus trichloride

Uses of phosphorus trichloride:

Phosphorus trichloride is used as a chlorinating agent and for the preparation of H_3PO_3.

Phosphorous pentachloride:

Preparation

When PCl_3 is treated with excess chlorine, phosphorous pentachloride is obtained.

$$PCl_3 + Cl_2 \longrightarrow PCl_5$$

Chemical properties

One heating phosphorous pentachloride, it decomposes into phosphorus trichloride and chlorine.

$$PCl_3\ (g) + \underset{\text{(Excess)}}{Cl_2(g)} \longrightarrow PCl_5(g).$$

Phosphorous pentachloride reacts with water to give phosphoryl chloride and orthophosphoric acid.

$$PCl_5 + H_2O \longrightarrow POCl_3 + 2HCl$$

$$POCl_3 + 3H_2O \longrightarrow H_3PO_4 + 3HCl$$

Overall reaction

$$PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$$

Phosphorous petachloride reacts with metal to give metal chlorides. It also chlorinates organic compounds similar to phosphorus trichloride.

$$2Ag + 2PCl_5 \longrightarrow 2AgCl + 2PCl_3$$

$$Sn + 4PCl_5 \longrightarrow SnCl_4 + 4PCl_3$$

$$C_2H_5OH + PCl_5 \longrightarrow C_2H_5Cl + HCl + POCl_3$$

$$C_2H_5COOH + PCl_5 \longrightarrow C_2H_5COCl + HCl + POCl_3$$

Uses of phosphorus pentachloride

Phosphorous pentachloride is a chlorinating agent and is useful for replacing hydroxyl groups by chlorine atom.

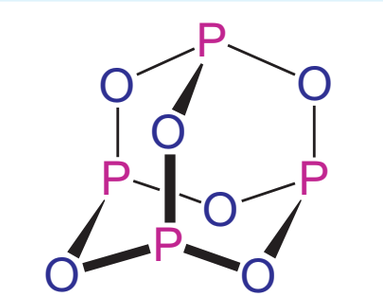


Figure 3.6 Structure of P_2O_3

3.1.11 Structure of oxides and oxoacids of phosphorus

Phosphorous forms phosphorous trioxide, phosphorous tetra oxide and phosphorous pentaoxides

In phosphorous trioxide four phosphorous atoms lie at the corners of a tetrahedron and six oxygen atoms along the edges. The P-O bond distance is 165.6

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